The Induction Principle — problem 8.17

A. Engel — Problem-Solving Strategies · 1998

For all $n \in \mathbb{N}$, we have $f(n) = g(n)$, where

$$f(n) = 1 - \frac{1}{2} + \frac{1}{3} - \cdots + \frac{1}{2n+1} - \frac{1}{2n}, \qquad g(n) = \frac{1}{n+1} + \cdots + \frac{1}{2n}.$$